ATHARVA GAUR

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EDUCATION

B.Tech, Computer Science Engineering (AI & ML) | VIT Vellore *Aug 2024 – May 2028*

CGPA: 9.53/10 | Coursework: DSA, OOP, OS, DBMS, Computer Networks, Linear Algebra, Discrete Mathematics

Class XII, Science (CBSE, 92.4%) | Air Force School, Subroto Park *2024*

TECHNICAL SKILLS

- **Languages:** Java, Python, JavaScript, TypeScript, C++, C, SQL (MySQL)
- **CS Fundamentals:** Object-Oriented Programming & Design, Data Structures & Algorithms, DBMS, Operating Systems, Computer Networks, Testing
- **Frontend:** React.js, Next.js, Tailwind CSS, Vite, React Router, Recharts, Axios
- **Backend & Databases:** Node.js, Express.js, FastAPI, REST API Design, MongoDB (Mongoose), MySQL, Zod
- **AI / GenAI:** LLM Integration (OpenRouter, Groq), RAG, Embeddings, Vector Databases (ChromaDB), LangChain, MCP, Prompt Engineering
- **ML & Data:** Scikit-learn, XGBoost, LightGBM, TensorFlow/Keras, NumPy, Pandas, SHAP
- **Tools & Cloud:** Git/GitHub, CI/CD (Vercel, Render), Linux, AWS (EC2, S3, Elastic Load Balancing, Auto Scaling)

PROJECTS

RAG-Based Intelligent Learning Assistant | LangChain, ChromaDB, FastAPI, React *Live* / *GitHub*

- Built a modular RAG pipeline for PDF question answering: semantic chunking, BGE-small embeddings (FastEmbed), ChromaDB vector retrieval and Groq-hosted LLMs, with prompts restricting answers to the uploaded document and page-level source citations.
- Implemented history-aware retrieval that rewrites follow-up questions into standalone queries for multi-turn conversations.
- Optimized for a 512 MB host with batched embedding and insertion and capped session memory; deployed with FastAPI on Render and React on Vercel, auto-redeploying on every GitHub push.

CareTrack | Full-Stack Issue Tracker | React, Node.js, Express, MongoDB, LLM API *Live* / *GitHub*

- Built an end-to-end issue tracker (React, Express, MongoDB) with REST APIs for creating, updating, filtering and assigning issues, a Kanban workflow, and Recharts status/priority dashboards.
- Integrated LLM-generated issue summaries and suggested actions via OpenRouter on synthetic healthcare operations data, with Zod-validated requests.
- Designed, built and deployed solo: React on Vercel, Express API on Render, database on MongoDB Atlas.

OPEN-SOURCE CONTRIBUTIONS

LangChain DeepAgents (Daytona Integration) | Python, HTTP Timeouts, pytest, Git *Merged PR #7121*

- Implemented Daytona request timeouts and preserved partial output from timed-out commands.
- Added SDK-level tests covering command execution, polling, log retrieval, and bounded cleanup.

LangSmith SDK (LangChain) | Python, Async Programming, Git *PR #63*, *PR #46*

- Fixed streamed Gemini traces in wrap_gemini dropping tool calls and finish_reason (Fixes #43), so traced LLM runs capture complete outputs.
- Added update_example to AsyncClient for sync/async API parity (part of #33), the first of a planned series of small PRs agreed with the issue author.

DSPy (Stanford NLP) | Python, MCP, pytest *PR #10642*

- Validated a fix for information loss in MCP tool-result conversion by building a live stdio-transport MCP integration test; ran the 34-test suite and flagged a docs mismatch the author fixed.

RESEARCH

Cross-Dataset Generalization in ML-Based DDoS Detection *Research Paper*

- Built a reproducible pipeline (Random Forest, XGBoost, LightGBM, Isolation Forest) across 3 CIC datasets; cut 77 features to 20 via mutual information.
- Showed performance degradation under distribution shift using frozen-model cross-dataset evaluation, SHAP and five-seed analysis.

COMPETITIVE PROGRAMMING & HACKATHONS

- 250+ problems solved on LeetCode, CodeChef and Codeforces (C++).
- Smart India Hackathon + inter-college hackathon (teams of 5-6): built the LLM chatbot for a crop price prediction system.